OmniFrame research: deprecated / over-thought tools, basics fixed

audit of omniframe mode + tool registry, 2026-10-22

1. Method

Every omniframe-mode control was asked one question: "can another, already-shipped tool do this exactly?" If yes, the control is a duplicate and was deleted (UI + invite), keeping ONE universal path. What survived must be a skill no other tool has. The headless MCP layer and the booted app now share the same maths modules (src/lib/selectionMattes.ts, src/lib/patchUnder.ts), so deleting UI never deletes capability.

2. Deleted from omniframe mode (and why each is a redo)

Duplicate character button (+ its panel invite)
Copy-paste is not an edit. It redoes duplicate_paste, and using it as the "move" path is exactly the cheat the omniframe test forbids: the moved thing must be a cut-out of the ORIGINAL clip, not a copy of the plate.

Transform section (X/Y sliders, reset)
Redoes transform_set / edit_element. After extract, the cut-out IS a normal clip, so the universal transform tools + keyframes already move/scale/rotate it. Two transform UIs = two truths.

Color / recolor section (dye blends)
Redoes adjustment layers + blend modes (blend_set, effect_set). Recolor blends were a third colour path; the app already has per-clip effects and adjustment LUTs.

Scope section (all frames / section / 1 frame)
Redoes clip ranges + keyframe timing. A cut-out clip spans frames like any clip; "which frames does my edit hit" is answered by keys and trim, not a parallel scope engine. Patch-under applies to every frame by construction.

Per-character transform evaluation (scope-aware)
Dead once scope is gone; CurveEngine evaluates everything. Kept only as legacy data for old projects, no UI, no new code.

3. Kept, because nothing else does it

- Transparent selection (omniframe signature skill): draw a rect BIGGER than the element; the ignore colour (white bed) is skipped, only the target (black box) is selected. New: src/lib/selectionMattes.ts transparentMatte, tool transparent_select, panel button, bridge call.
- Mind-main-element: flood the background from the rect border; the survivor is the subject (mainElementMatte, select_main_element).
- Extract cut-out: one button/tool creates a NEW clip on the track above holding only the selected section of the source, all frames (omniframe_extract / store omniExtractCutout).
- Patch under: smart ring-average fill hides the hole on the source for every frame (patchUnder lib, playback.ts + appRender.mjs both call it).
- Roto sub-tool and segment detection stay: they are the entry points that produce selections.

4. Over-thought, rebuilt thin

Observation (user): extract+patch-under is "just masking / video-view-mask that clones instead of editing the original and shows the clone on top" - correct. So it is no longer a parallel engine: it composes universal primitives (inverted mask = hole, smart patch = fill, sprite asset = cut-out clip). ~200 lines of shared slice replace the old character-scope machinery. Video-view-mask itself stays as a RECIPE of universals (duplicate + mask include + mask exclude), not a tool.

5. Basics fixed while auditing

- Video encoder streamed to disk: holding 500+ raw RGB frames in RAM OOM-killed long exports (exit 137). Now frames stream to a temp .rgb and ffmpeg reads the file.
- Headless key values are absolute centre-offsets (CurveEngine getVal replaces base once keys exist); recipe geometry corrected, no more "box never travels".
- Custom curves: InterpolationType gained 'custom' with user-dragged normalized points (customProgress), in the app CurveEngine so UI and headless agree; recipe E ships 5 of them.
- playback.ts never applied clip.patch: patch-under now renders in the booted app, not only headless.
- Timeline snapshot ruler no longer collides with the title block.

6. New universal tools (registry 32 -> 40; bridge 15 -> 21)

clip_stretch (img stretches infinitely, rule), clip_remove, transparent_select, brush_select, select_main_element, omniframe_extract, patch_under, omniframe_move (constant step = no slide). Bridge mirrors 6 of them onto the real store; panel gains Transparent / Main element / Extract cut-out / Patch under buttons.

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7. Ideas parked (not built, on purpose)

- Learned subject matting behind transparent selection for busy backgrounds (today: colour-distance + flood, deterministic and testable).
- Curve-editor drag handles writing 'custom' point lists (maths exists; UI drag later).
- Per-frame patch refinement (optical-flow fill) - ring-average already passes the white/graduated beds; flow is weight we do not need yet.
- Compound pairing button for video-view-mask (duplicate+exclude+include in one click) - stays a recipe until users ask.

8. Evidence

test_ans_result/E_move/solution.mp4 (24s: 5 flash holds x2s with byte-identical frames, 15 curve segments incl. 5 custom, eased move to destination); evidence/E-snap-*.png (view-mask hole, extract+patch, holds, jump, destination); evidence/timeline-E.png (tracks, MASK badge, keys, log); qa/universal-tools-unit.mjs PASS (63 universal steps, 40 tools).